# CS2103 / Lab – Assignment 11

**Topic: Part-of-Speech Tagging using Hidden Markov Models**

**Instructions:** Complete all five questions. Implement the **HMM-based POS tagger and Viterbi decoding from scratch using Python/NumPy**. Built-in POS taggers or HMM libraries are not allowed except for verification.

**Q1. HMM Representation for POS Tagging**

Consider the Hindi sentence:

सीता बगीचे से सुंदर लाल फूल तोड़ती है।

with possible POS tags

$$\{NNP, NN, PSP, JJ, VM, VAUX\}.$$

Represent the words as **observations** and the POS tags as **hidden states**. Construct suitable transition and emission probability tables for the sentence and store them using appropriate Python data structures.

Display both probability tables clearly.

**Q2. Sequence Probability**

Consider the candidate POS sequence

$$NNP \to NN \to PSP \to JJ \to JJ \to NN \to VM \to VAUX.$$

Using the transition and emission probabilities constructed in Q1, compute programmatically the joint probability of this sequence.

Your implementation should multiply the required $P(t_i \mid t_{i-1})$ transition probabilities and $P(w_i \mid t_i)$ emission probabilities, including the START and END transitions.

Display all intermediate probability terms and the final sequence probability.

**Q3. Viterbi Algorithm from Scratch**

Implement the Viterbi algorithm to find the most probable POS-tag sequence for

सीता बगीचे से सुंदर लाल फूल तोड़ती है।

For each word position and candidate tag, compute the best probability using the previous state, transition probability, and emission probability.

Store the best predecessor for each state and use **backtracking** to recover the complete POS sequence.

Print the Viterbi probability table, backpointers, and final predicted tag sequence.

**Q4. Ambiguous POS Tagging**

Modify the emission table so that one word can have more than one possible POS tag. For example, allow

सुंदर

to have non-zero emission probabilities for two candidate tags.

Run Viterbi again and observe the selected tag. Change one relevant transition or emission probability and rerun the algorithm.

Explain how both **context** and **word likelihood** influence the final POS decision.

**Q5. Viterbi Analysis Challenge**

Using the same HMM implementation, construct and tag at least **three short Hindi sentences** using words from the available vocabulary.

For each sentence, print:

(i) the Viterbi probability table,
(ii) the backpointer information,
(iii) the final predicted POS sequence.

Compare the results and briefly explain how transition and emission probabilities jointly determine the most probable tag sequence.

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**Note:** Students must be able to explain observations, hidden states, transition probability, emission probability, sequence probability, Viterbi dynamic programming, backpointers, and POS ambiguity during the evaluation/viva.